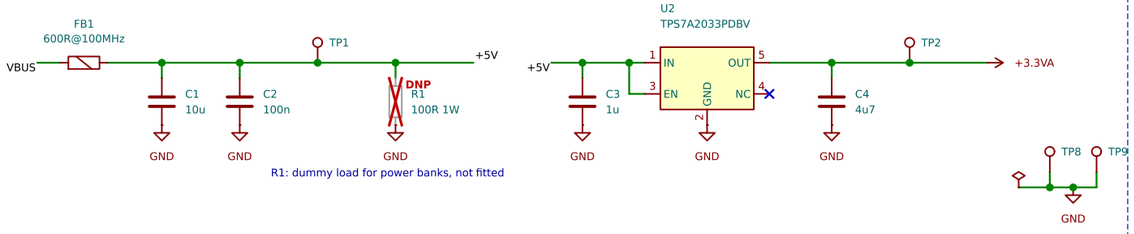
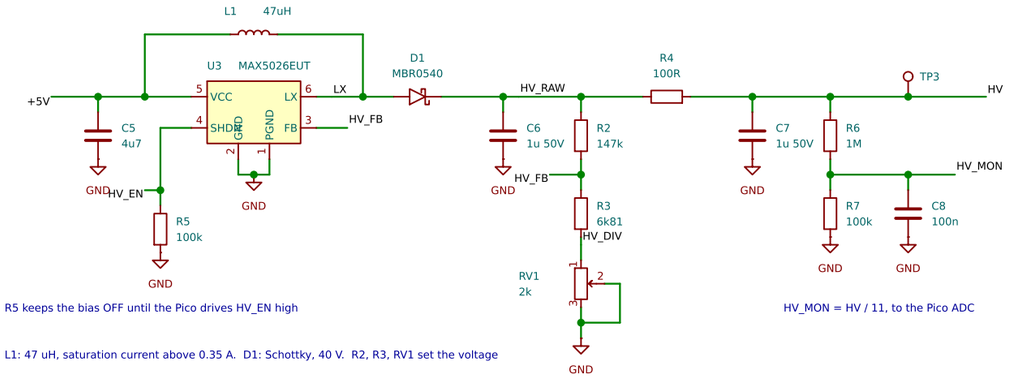
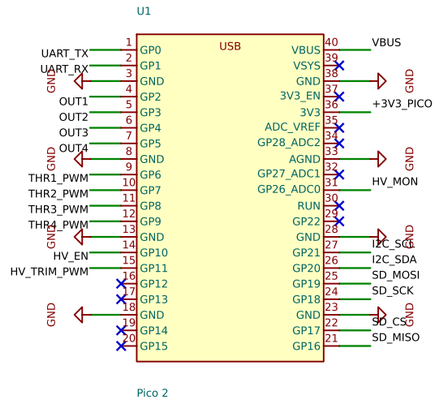
1. 5 V input and 3.3 V analog supply



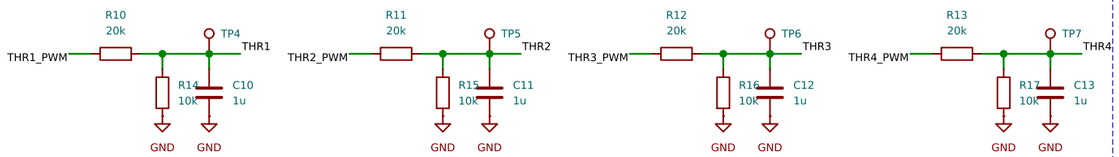
2. SiPM bias: MAX5026 step-up, 22 - 28 V



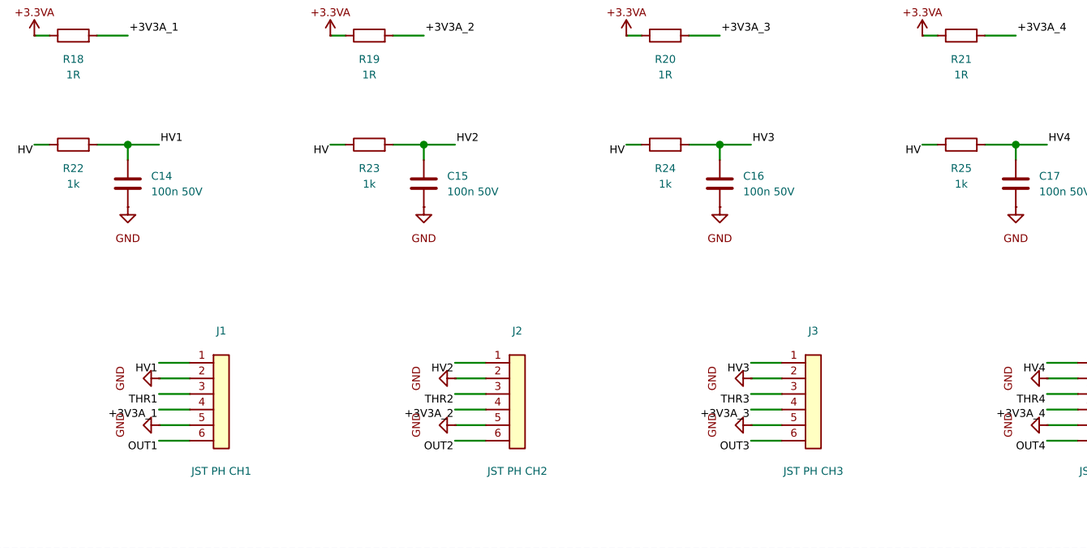
3. Raspberry Pi Pico 2



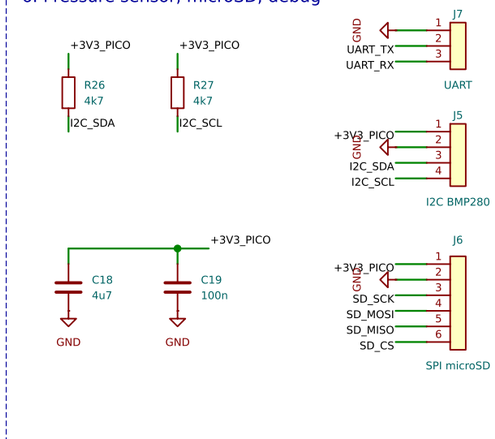
4. Thresholds: PWM / 3, filtered. 0 - 1.1 V



5. Channel connectors: 1 HV, 2 GND, 3 THR, 4 +3V3A, 5 GND, 6 OUT (straight cable, pin 1 to pin 1)



6. Pressure sensor, microSD, debug



PCB-402: Pico 2, SiPM bias, analog 3.3 V, four thresholds, four channel connectors
Bias 22.1-28.2 V set with RV1; R1 and R8 not fitted

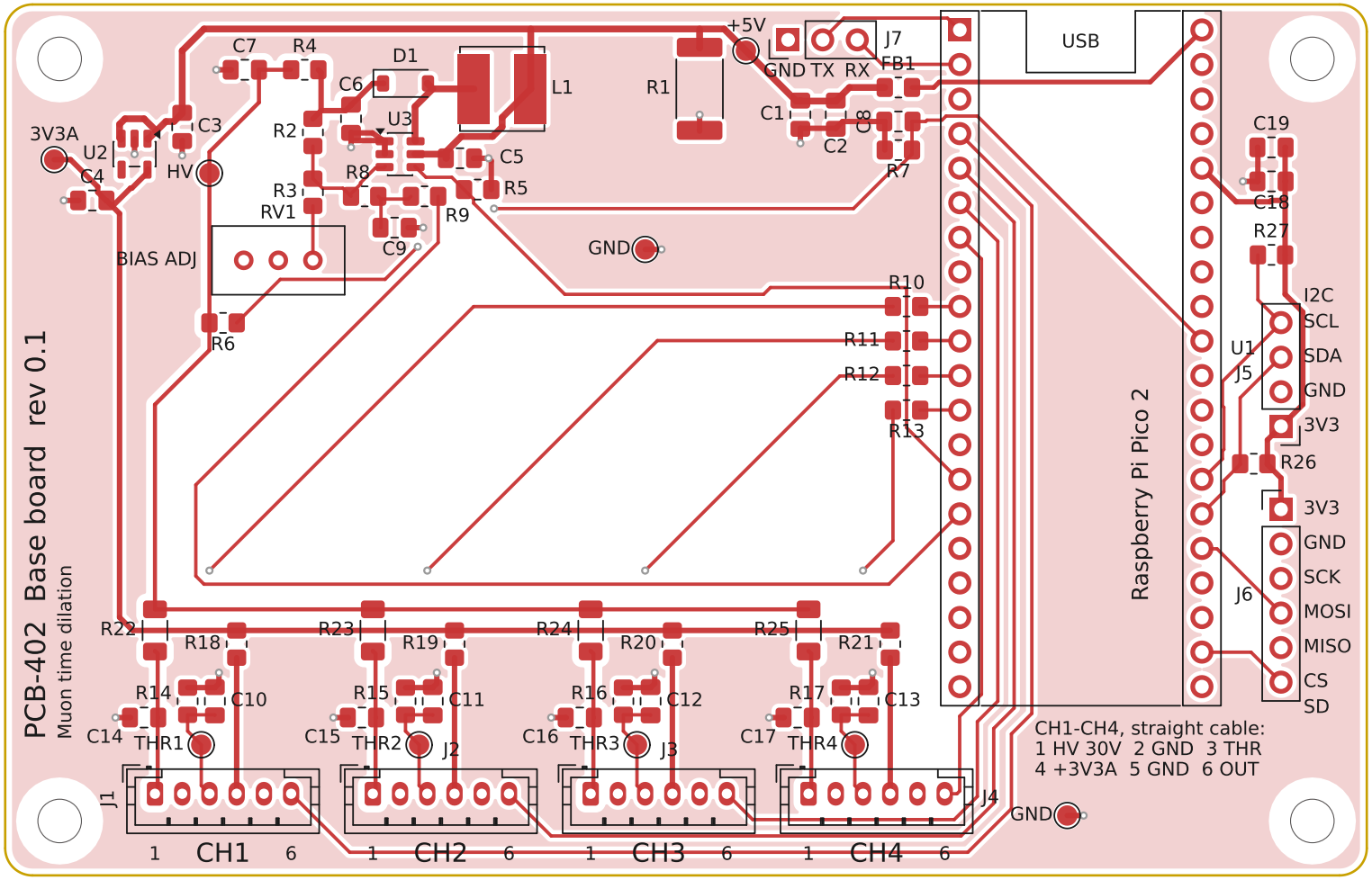
Muon time dilation project

Title: Base board

Size: A3 Date: 2026-10-11 Rev: 0.1 File: PCB-402_base_board.kicad_sch

PCB-402: top view (copper, silkscreen, outline)

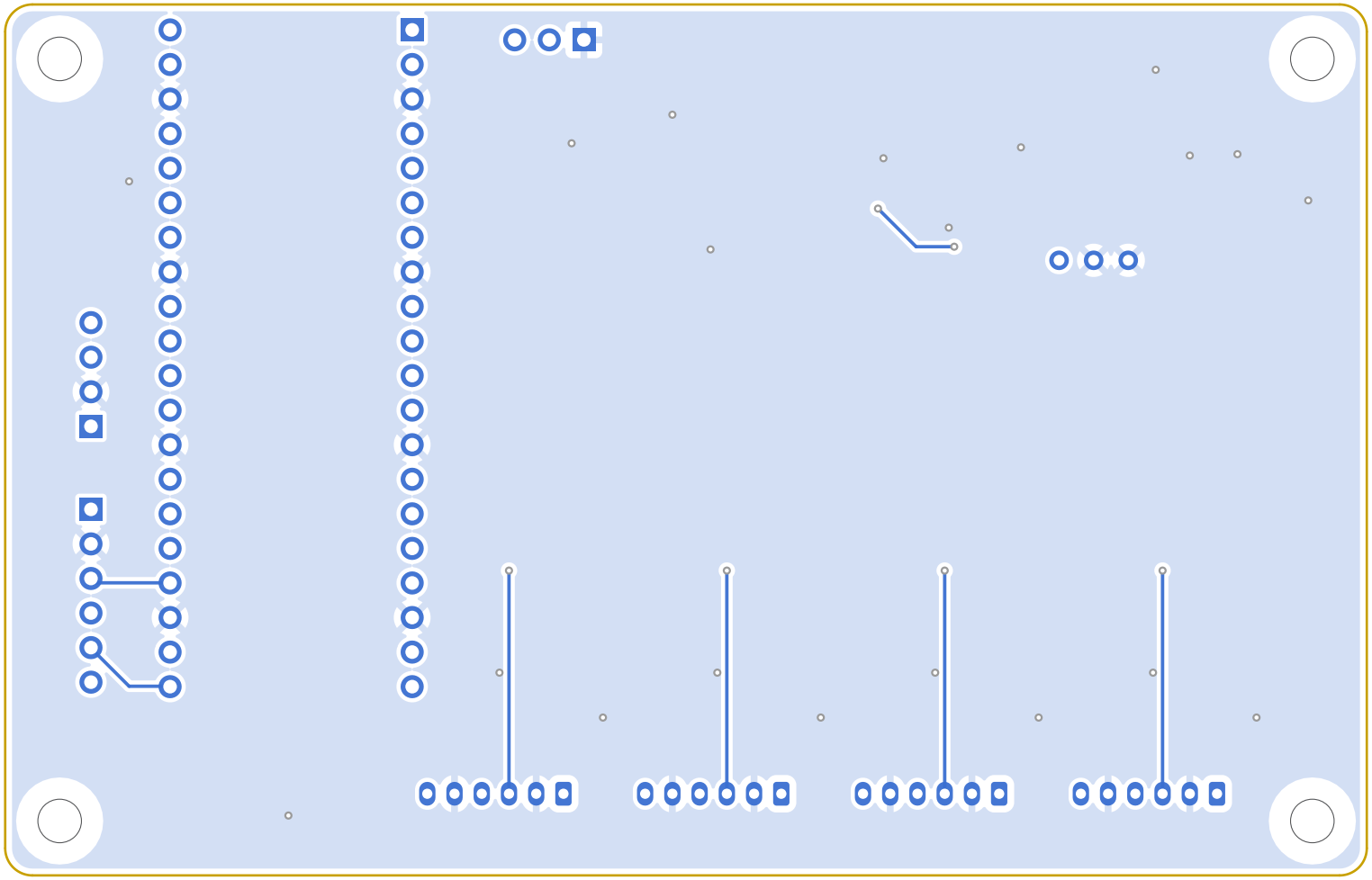
PCB-402_base_board.kicad_pcb · board 100.0 x 64.0 mm · scale 2:1 on A4 · rev 0.1



10 mm

PCB-402: bottom view, mirrored (as seen when the board is flipped)

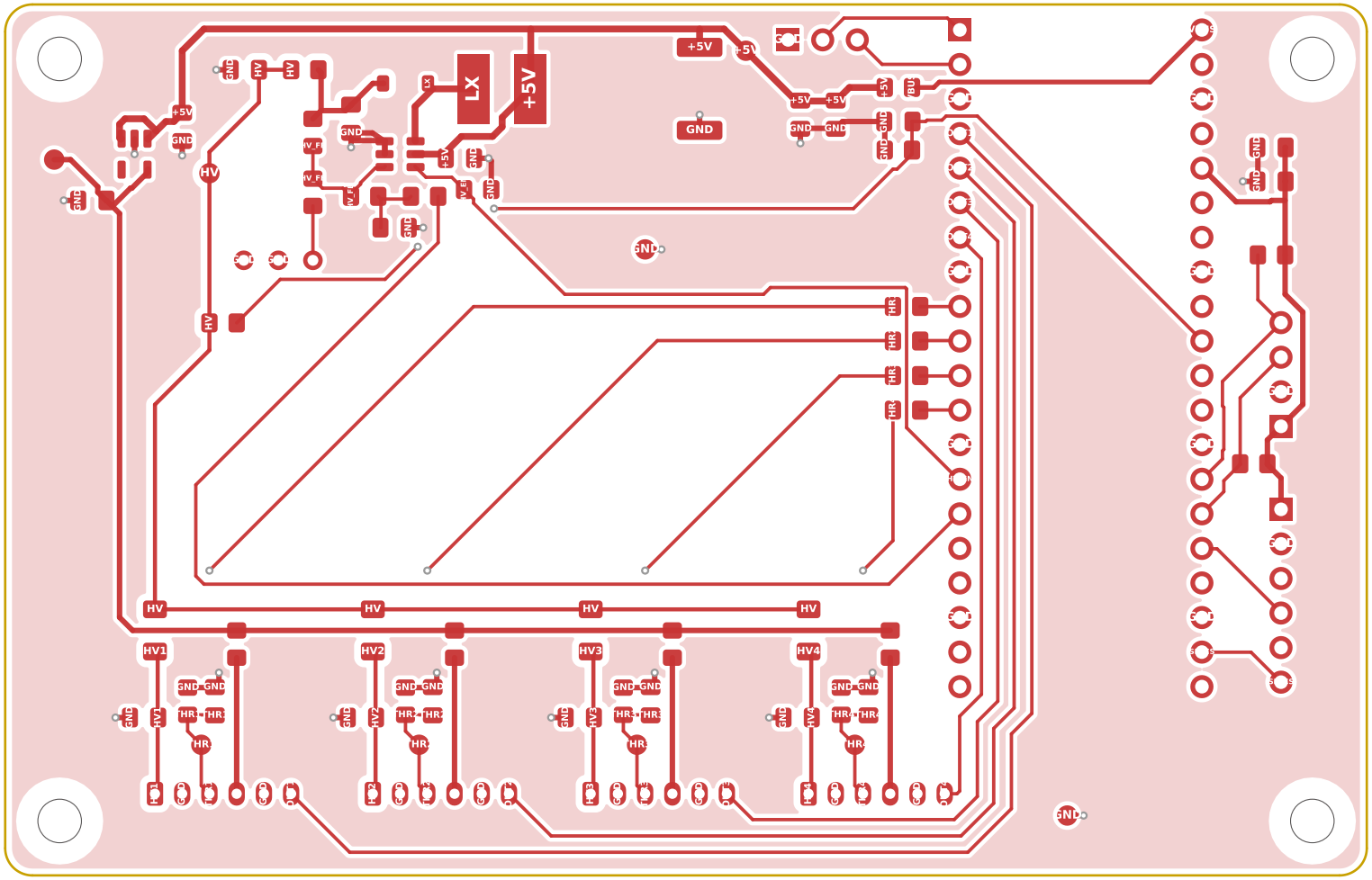
PCB-402_base_board.kicad_pcb · board 100.0 x 64.0 mm · scale 2:1 on A4 · rev 0.1



10 mm

PCB-402: top copper (F.Cu) with net names on the pads

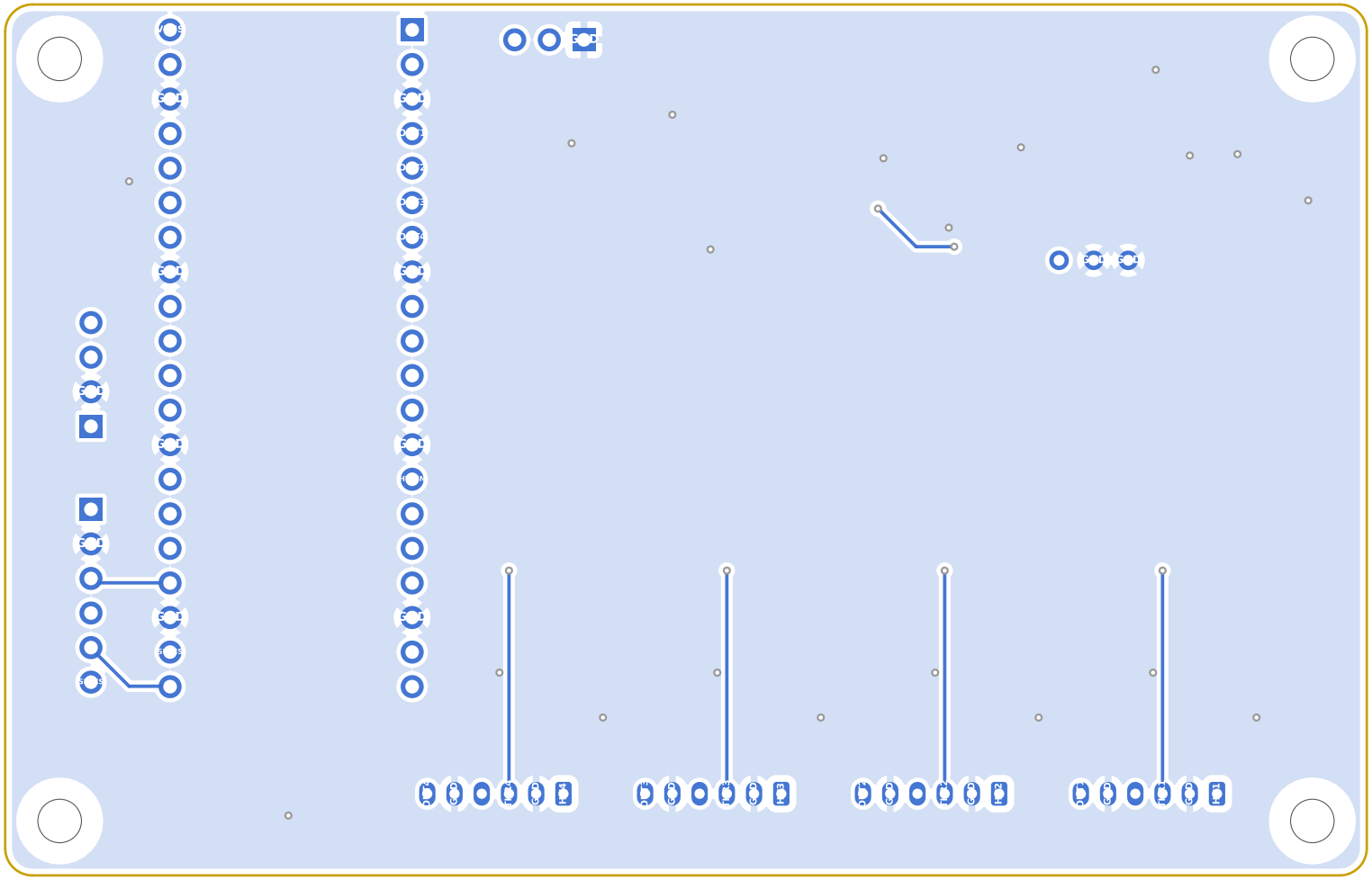
PCB-402_base_board.kicad_pcb · board 100.0 x 64.0 mm · scale 2:1 on A4 · rev 0.1



10 mm

PCB-402: bottom copper (B.Cu), mirrored, with net names on the pads

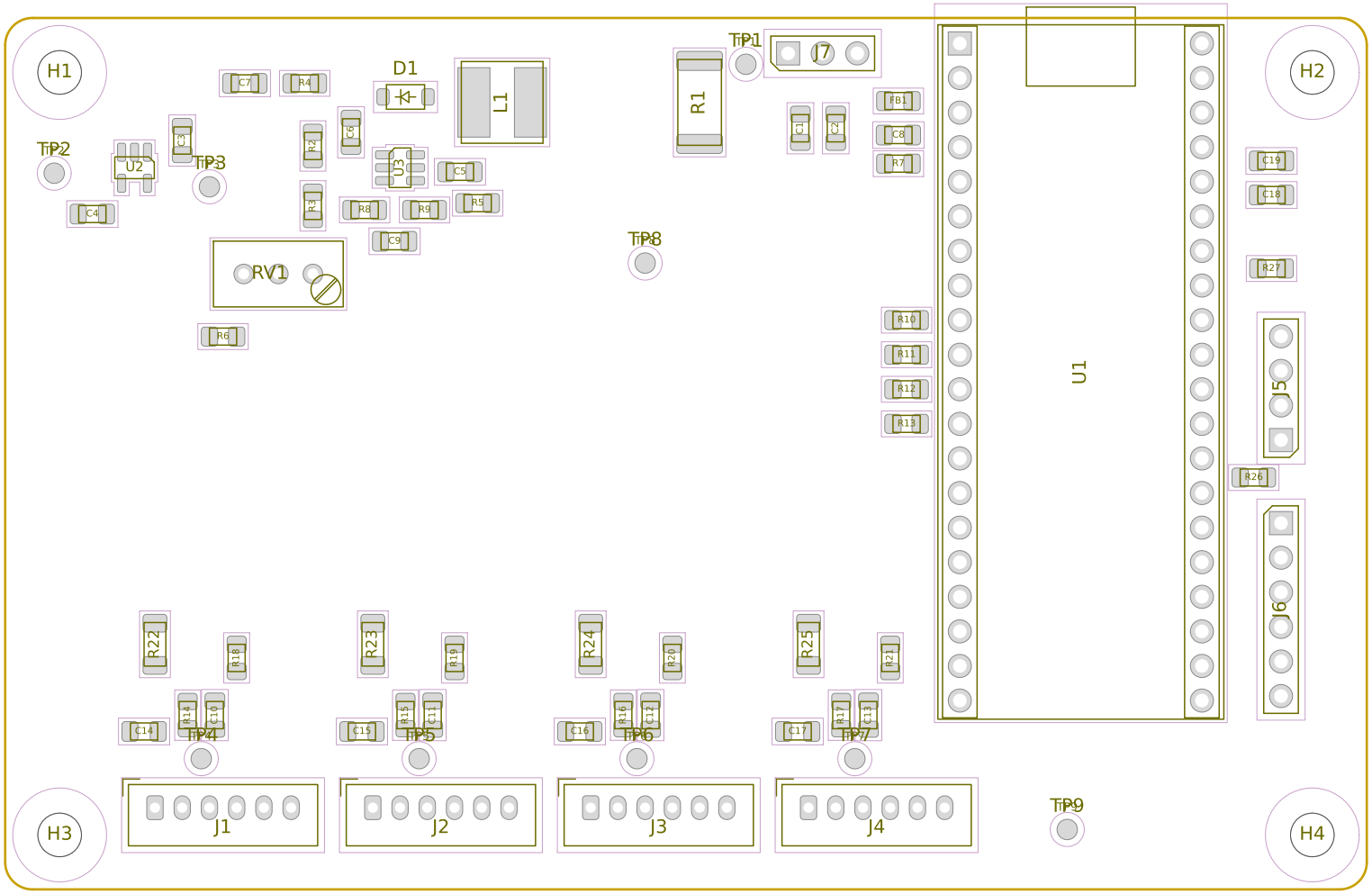
PCB-402_base_board.kicad_pcb · board 100.0 x 64.0 mm · scale 2:1 on A4 · rev 0.1



10 mm

PCB-402: top-side assembly (references and component outlines)

PCB-402_base_board.kicad_pcb · board 100.0 x 64.0 mm · scale 2:1 on A4 · rev 0.1



10 mm